

**Erica F. Huang** Wellesley, MA | (781) 985-0004 | erica.f.huang@gmail.com | <https://github.com/erica-f-huang>

## EDUCATION

**Tufts University**, Medford, MA

May 2027

B.S.E. in Computer Science, Minor in Embedded Systems

3.88/4.0 GPA, Dean's List (5 semesters)

**Relevant Courses:** Data Structures, AI: Algorithms, Ethics, & Policy, Machine Structure & Assembly Language Programming, Embedded Systems, Human Computer Interaction, Digital Circuits, Technical Communication, Security

## EXPERIENCE

**Research Assistant (Paid), Tufts University Human Robot Interaction Lab**, Medford, MA

May 2025 – Present

- Training Boston Dynamics' Spot robot to improve object manipulation skills in an open-world environment
- Running experiments to understand how teams work to create an LLM to assist teamwork

**Software Engineering Intern (Paid), Cytel**, Wellesley, MA

Summer 2024, Summer 2025

- Optimized code using parallel processing and communicating between R and C++ for faster runtime
- Generated reports using simulation output in R to provide clients with a formatted summary of their simulated trials
- Tested different forms of optimization (using a dynamically linked library to communicate between C++ and R, running C++ in parallel, and running R in parallel) to understand which method results in the fastest runtime
- Presented well-received analysis and recommendations to a team of senior developers for further development of my work

**Founder and Lead Student Mentor, FIRST LEGO League (FLL) Team**, Wellesley, MA

May 2022 – May 2023

- Created and led a partnership with Wellesley Middle School science and technology teachers to launch two FIRST LEGO League Challenge teams for 6th through 8th-grade students
- Placed 9th and 19th out of 36 teams at the 2022 regional competition
- Won the team award for displaying FIRST core values
- Currently a team advisor and mentor during university breaks for around 20 students

**Founding Member and Coding Lead, FIRST Robotics Competition (FRC)**, Wellesley, MA

Aug 2020 – May 2022

- Led a nine-member coding team, contributing to an award-winning robot
- Expanded and gender-diversified coding team, contributing to a second award-winning robot
- Established and served in the team's first Director of Community Education and Outreach role
- Created and implemented summer robotics workshops for 3rd and 4th graders and collaborated with local Girl Scout leaders to develop and run badge-earning events

## AWARDS AND ACHIEVEMENTS

**Centennial Youth Service Award**

2023

Awarded by the town of Wellesley for founding and leading two FIRST LEGO League teams for the town's public middle school.

**Dean's List Nominee, FIRST Robotics Competition**

2022

Nominated for leadership and professionalism by my 40-member team and mentors from the community and industry.

## SKILLS, INTERESTS, HOBBIES, EXTRACURRICULARS

**Skills:** C++, C, Java, R, Python, LaTeX, robotics, human-robot interaction, parallel computing, optimization, neural networks, digital circuits, digital electronics, embedded systems, human-computer interaction, Linux, Git

**Interests:** Robotics, human-robot interaction, artificial intelligence, programming, machine learning, mechanical and electrical systems design, physics, calculus, performance optimization, ethical applications of technology, embedded systems

**Hobbies:** Snowboarding, weightlifting, basketball, triathlons, rock climbing, backpacking, puzzles, saxophone

**Tufts Clubs:** Basketball, Wilderness Orientation, Dance Collective, Outreach Learning Fellows, Robotics